Skills Summary

Rates, Extrema, and the Shape of a Graph

Use this reference while completing your worksheet or when studying.

Skill 1 — Related Rates: Differentiate First

Tell: the problem gives a rate and asks for a rate.

Write the geometric relation, differentiate every variable with respect to t, and substitute the instant's numbers only afterwards: $\dfrac{d}{dt}\big(s^n\big) = ns^{n-1}\dfrac{ds}{dt}$.

Example: A cube's edge grows at 2 cm/s. Find $\dfrac{dV}{dt}$ when $s = 3$ cm.

Step 1. A rate is given and a rate is wanted, so differentiate the volume relation with respect to time rather than solving for a number first.

Step 2. $V = s^3$, so $\dfrac{dV}{dt} = 3s^2\dfrac{ds}{dt} = 3(3)^2(2)$

$\Rightarrow$ $\boldsymbol{\frac{dV}{dt} = 54}$ cubic centimetres per second.

Try it: Same cube, same 2 cm/s. Find $\dfrac{dV}{dt}$ when $s = 5$ cm.

check: **150**

Skill 2 — Optimization: Constraint and Objective

Tell: the problem says largest, smallest, least, or most.

Name the objective (what is optimized) and the constraint (what is fixed), use the constraint to reduce the objective to one variable, then set the *derivative* to zero — never the objective itself.

Example: A rectangle's perimeter is 60, so $A(x) = x(30 - x)$. Maximize the area.

Step 1. The area is already in one variable, so the only remaining move is to differentiate it and set that derivative to zero.

Step 2. $A(x) = 30x - x^2$, so $A'(x) = 30 - 2x$, and $30 - 2x = 0$ gives the critical value.

$\Rightarrow$ $\boldsymbol{A'(x) = 30 - 2x}$, maximum at $\boldsymbol{x = 15}$.

Try it: Same setup with $A(x) = x(36 - x)$: find $A'(x)$ and the maximizing x.

check: $\boldsymbol{A'(x) = 36 - 2x,\ x = 18}$

Skill 3 — Curve Analysis from f' and f''

Tell: the problem asks where a graph rises, turns, or bends.

$f' = 0$ locates critical points; the sign of f' says rising or falling. $f'' = 0$ *with a sign change* locates inflection points; the sign of f'' says concave up or down.

Example: For $f(x) = x^3 - 3x^2 + 1$, find $f'(x)$ and the critical values.
Step 1. Critical points are where the first derivative vanishes, so differentiate once and solve that equation.
Step 2. $f'(x) = 3x^2 - 6x = 3x(x-2)$, which is zero at each factor's root.
$\Rightarrow$ $\boldsymbol{f'(x) = 3x^2 - 6x}$, critical values $\boldsymbol{x = 0}$ or **2**.

Try it: For $f(x) = x^3 - 6x^2 + 2$, find $f'(x)$ and the critical values.

check: $\boldsymbol{f'(x) = 3x^2 - 12x}$, $\boldsymbol{x = 0}$ or **4**